VENKATA GURAVAREDDY

PRINCIPAL AI & PLATFORM INFRASTRUCTURE ARCHITECT

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EXECUTIVE SUMMARY

Principal AI & Platform Infrastructure Architect with 14 years architecting, provisioning, and operating enterprise-scale distributed compute platforms, cloud abstractions, and high-density GPU supercomputers. Deep subject matter expert spanning the full stack from physical bare metal hardware lifecycle (PXE, IPMI, BIOS, PCIe) to GPU inference serving platforms (NVIDIA H100/L40S, vLLM, Triton, InfiniBand, RDMA, NVLink) and Kubernetes control planes. Proven track record managing 600+ Kubernetes/OpenStack clusters across 52 data centers, scaling edge container platforms to 42,000+ nodes, and engineering distributed telemetry and capacity engines processing 5M+ records/hour ($1.8M/yr spend reduction). Granted US Patent Holder (US 12,028,213 B2 & US 11,336,525 B1) and recipient of 20+ Verizon Spotlight Awards for sustained technical leadership and platform excellence.

CORE TECHNICAL COMPETENCIES

AI, GPU & Serving Runtimes:

NVIDIA H100, L40S, A100, DGX/HGX, vLLM, Triton Inference Server, LiteLLM Proxy, Continuous Batching, Dynamic GPU Slicing, KV-Cache Optimization, CUDA, GPU Operator, DCGM Telemetry, Fabric Manager.

Platform, Cluster & Bare Metal:

Kubernetes (Core & Custom Controllers), Red Hat OpenShift, OpenStack, KVM, Bare Metal Provisioning, IPMI/Redfish, PXE Boot, BIOS & Kernel Tuning, Linux (RHEL/Ubuntu), Multi-Tenant Admission Control.

High-Speed Fabric & Networking:

InfiniBand NDR/XDR, RoCEv2/RDMA, NVLink/NVSwitch Interconnects, TCP/IP, SR-IOV, DPDK, CNI (Calico/Cilium/Multus), BGP, VXLAN, Load Balancing (Envoy/F5), Cross-Cluster Routing, TLS/mTLS, eBPF.

Distributed Storage, Automation & SRE:

Ceph, MinIO S3, NetApp Trident (CSI), Python, Go, Bash, Terraform, Ansible, ArgoCD, Helm, GitOps, Prometheus, Grafana, ELK Stack, Distributed Capacity Analytics, Sev-0 Incident Commander.

PROFESSIONAL EXPERIENCE

Verizon | *Principal Engineer : AI & Platform Infrastructure*

2017 - Present | Hyderabad, India

Lead Architect : Enterprise GPU Infrastructure & GenAI Inference Serving Platform

2023 - Present

- End-to-End GPU Architecture: Architected Verizon's centralized GPU inference compute platform from physical bare metal to model serving; provisioned and tuned multi-node NVIDIA H100 and L40S clusters with GPU Operator, DCGM, and Fabric Manager.
- High-Throughput Serving Pipelines: Integrated vLLM, LiteLLM proxy, and Triton Inference Server with dynamic GPU slicing, continuous batching, and KV-cache optimization, boosting model throughput by 3.8x while meeting strict P99 latency SLOs.
- High-Speed Lossless Fabric & SRE: Configured InfiniBand/RDMA fabric; resolved a live 94% NVLink saturation incident and PCIe bus errors using real-time DCGM profiling, instituting full-stack observability achieving 99.99% infrastructure uptime.

Staff Platform Engineer : Fleet Kubernetes & Distributed Capacity Automation

2021 - 2024

- Planet-Scale Fleet Operations: Owned and operated 600+ Kubernetes and OpenStack production clusters across 52 data center sites, standardizing bare metal lifecycle, CNI overlay networks, and persistent storage via Ceph and NetApp Trident CSI.
- Distributed Capacity Analytics Engine (CER): Engineered CER, a distributed telemetry and capacity forecasting engine in Python processing 5M+ records/hour across 500+ clusters and 60,000+ pods, saving $1.8M annually in compute over-provisioning.
- Multi-Tenant Governance & Automation: Enforced namespace quotas, pod disruption budgets, custom admission webhooks, and network policies; developed Python, Go, and Terraform toolchains for automated node lifecycle, OS patching, and self-healing.

PROFESSIONAL EXPERIENCE (CONTINUED)

Verizon (Continued) 2017 - 2021

Platform Lead : Atlas 5G Cloud Native Network Platform 2017 - 2021

- Scaled Platform to 42,000+ Edge Nodes: Led the architectural design and implementation of Atlas (5G CNF deployment, verification, and admission control engine); scaled from a 2-person prototype to 42,000+ edge locations and 10,000 operations/hour.
- Granted Patents & Vendor Disruption: Co-invented core pre-flight container validation and policy admission architecture (Patents US 12,028,213 B2 & US 11,336,525 B1), displacing legacy commercial software (HP NFVD) corporate-wide.
- Team Leadership & Reliability: Scaled and mentored cross-functional team from 2 to 14 engineers; established GitOps CI/CD pipelines, automated canary gates, and operational runbooks delivering 99.999% platform availability across nationwide nodes.

PREVIOUS CAREER EXPERIENCE

Ericsson R&D | *Senior Software Engineer* 2016 - 2017 | Bangalore, India

- Built modular Python microservices and CI/CD validation harnesses for Virtualized Network Function (VNF) lifecycle orchestration integrated with OpenStack and SDN controllers.
- Contributed upstream network virtualization automation code to open-source telecommunications and 5G working groups.

Hewlett Packard Enterprise (HPE R&D) | *Software Engineer* 2014 - 2016 | Bangalore, India

- Automated bare metal server provisioning, BIOS injection, and configuration management for HPE ProLiant Gen8/Gen9 rack servers using Python, IPMI, and OpenStack Cinder/Ironic.
- Selected for on-site US customer deployment; led physical datacenter rack bring-up, SAN storage connectivity, and operational handoff.

HCL Technologies (Cisco Practice) | *Software Engineer* 2012 - 2014 | Chennai, India

- Authored automated Python regression suites for Cisco IOS/IOS-XE firmware; investigated and resolved customer-escalated production defects in partnership with Cisco TAC.
- Configured physical multi-hop router and switch lab topologies replicating customer routing and high-availability failovers.

PATENTS & INTELLECTUAL PROPERTY (USPTO)

GRANTED **US 12,028,213 B2:** Systems and Methods for Validating a Container Network Function (CNF) for Deployment (Granted July 2, 2024 · Assignee: Verizon Patent and Licensing Inc.).

GRANTED **US 11,336,525 B1:** Automated Containerized Network Function Validation, Security Profiling, and Policy-Enforced Deployment System (Granted May 17, 2022 · Assignee: Verizon Patent and Licensing Inc.).

HONORS & KEY RECOGNITIONS

★ **20+ Verizon Spotlight Awards**
Recognized for technical leadership, Sev-0 incident resolution, GPU platform bring-up, and engineering impact.

★ **Academic Gold Medalist**
Graduated Department Valedictorian with First Rank across University in Computer Science & Engineering (Andhra University, 2012).

EDUCATION

B.Tech : Computer Science & Engineering
Andhra University | 2012
Department Gold Medalist (First Rank across University)

CERTIFICATIONS

✓ Red Hat Certified Specialist in OpenShift Administration
✓ Cisco Certified Network Associate (CCNA)